

Indian Railways

A to Z

History, trains, categories and facts

From the first 1837 goods line to Vande Bharat:
zones, gauges, train types, numbering,
coach classes, quotas, records and a glossary.

Contents

1. Indian Railways at a glance
2. Timeline: 1832 to today
3. Railway gauges
4. Zones and headquarters
5. Train categories (premium to local)
6. Luxury, tourist and special trains
7. Train numbering system
8. Coach classes
9. Reservation quotas and booking status
10. Heritage railways and metros
11. Records and landmarks
12. A to Z of notable trains
13. A to Z glossary
14. Sources and how to verify

How to use this guide. It moves from the big picture (network, zones) down to the individual train names and coach codes a passenger meets on a ticket. Dates and figures come from widely published sources and are rounded; where a number changes often it is marked *about*.

1. Indian Railways at a glance

Indian Railways is the state-owned railway system of India, run by the Ministry of Railways from Rail Bhavan, New Delhi. It is one of the largest rail networks in the world and the backbone of long-distance travel and freight in the country.

Item	Detail (approximate)
Owner / operator	Government of India, Ministry of Railways; policy body is the Railway Board
Route length	About 68,000 route km, with a running-track length well above that
Stations	About 7,000+ stations
Zones	17 operational zones (16 regional zones plus Kolkata Metro); an 18th, South Coast Railway (Visakhapatnam), was announced in 2019
Passengers	Billions of passenger journeys a year, including suburban traffic
Freight	More than 1.5 billion tonnes a year; coal, iron ore, cement, food grains and containers lead
Workforce	Roughly 1.2 to 1.3 million employees
Main gauge	Broad gauge, 1,676 mm (5 ft 6 in)
Electrification	Most of the broad-gauge network is electrified at 25 kV AC
Booking	IRCTC (founded 1999) runs online ticketing, catering and tourism

Indian Railways is commonly ranked among the four largest railway networks in the world by route length, alongside the United States, China and Russia.

2. Timeline: 1832 to today

Period	What happened
1832 to 1837	First proposals for railways in India. In 1837 the Red Hill Railway near Madras (Chennai) carried granite for road building, generally treated as the first train run in India (a goods line).
1845 to 1851	Short work-site lines: Godavari Dam Construction Railway (1845) and the Solani Aqueduct line at Roorkee (1851), used to haul construction material.
1853	Lord Dalhousie's Railway Minute lays out a national plan. On 16 April 1853 the first passenger train ran from Bori Bunder (Bombay) to Thane, about 34 km, on the Great Indian Peninsula Railway (GIPR).
1854 to 1856	East Indian Railway runs Howrah to Hooghly (1854). Madras Railway opens Royapuram to Wallajah Road (1856).
1860s to 1870s	Companies build trunk lines under a guarantee system. Bombay and Calcutta are linked (1870); Delhi and Calcutta by rail (1860s); the network passes several thousand miles.
1880 to 1900	Hill railways appear: Darjeeling Himalayan Railway (1881). Metre-gauge and narrow-gauge feeders spread across princely states and provinces.
1905	Railway Board created (Curzon era) to coordinate the companies; state purchase of company lines gathers pace afterwards.
1907 to 1914	Matheran Hill Railway (1907), Nilgiri Mountain Railway (1908), Kalka to Shimla (1903), and the original Pamban Bridge to Rameswaram (1914).
1912 to 1930	Punjab Mail (1912). A separate Railway Budget begins in 1924. First electric train in India in 1925 on Bombay's Harbour line (1,500 V DC). Frontier Mail (1928) and Deccan Queen (1930) become famous.
1947	Independence and Partition: a large part of the network, including lines and workshops, goes to Pakistan; the Indian side is reorganised.
1951 to 1958	Princely and company lines are merged and nationalised. Regional zones are created: Southern, Central and Western (1951); Northern, Eastern and North Eastern (1952); South Eastern (1955); Northeast Frontier (1958). AC traction on 25 kV is adopted from 1957.
1960s to 1970s	Diesel locomotives replace steam on many routes. Rajdhani Express (1969) links New Delhi and Howrah. South Central Railway zone formed (1966).
1980s	Palace on Wheels (1982); Kolkata Metro opens in 1984, India's first metro; computerised reservations begin at New Delhi (1986); Shatabdi Express introduced (1988).
1990s	Project Unigauge converts metre and narrow lines to broad gauge (from 1992). Steam is retired from regular service. Konkan Railway opens fully in 1998. IRCTC formed in 1999.
2002 to 2010	Six new zones announced (operational from 2003). IRCTC online booking (2002). Garib Rath (2006), Durgam Cheruvu (2009), Maharajas' Express (2010). Kolkata Metro becomes a zone (2010).
2014 to 2019	Gatimaan Express reaches 160 km/h (2016). Humsafar, Antyodaya, Tejas and Uday trains added. Railway Budget merged into the Union Budget (2017). Vande Bharat Express launched in February 2019 between New Delhi and Varanasi.
2020 to today	Kavach, an indigenous train-protection system, is deployed on selected routes. Amrit Bharat non-AC trains (from Dec 2023), Vande Bharat expansion including sleeper and chair-car versions, Vande Metro and Namoh Bharat rapid rail services, the new Pamban Bridge and the Chenab Arch Bridge on the Kashmir link (2025).

The 1837 to 1851 items were work-site or goods lines; the 1853 Bombay to Thane run is treated as the start of public passenger railways in India.

3. Railway gauges

Gauge is the distance between the inner faces of the two rails. India inherited several gauges and has spent decades unifying them.

Gauge	Width	Where it is used
Broad gauge (BG)	1,676 mm (5 ft 6 in)	Main lines, about nine-tenths of the network; all major trains
Metre gauge (MG)	1,000 mm	Once widespread; mostly converted under Project Unigauge; a few lines remain
Narrow gauge (NG)	762 mm (2 ft 6 in) and 610 mm (2 ft)	Hill and heritage lines: Darjeeling, Matheran, Kalka to Shimla, Nilgiri (metre), Kangra Valley
Standard gauge	1,435 mm	Some metros and urban systems, not the main railway network

Broad gauge gives stability at high speed and heavy loads, which is why the network is being concentrated on it. **Project Unigauge** (launched 1992) aimed for a single broad-gauge network.

4. Zones and headquarters

The network is split into zones, each headed by a General Manager and divided into divisions. Codes in brackets appear on tickets and timetables.

Zone	Headquarters	Formed
Central Railway (CR)	Mumbai (CSMT)	1951
Eastern Railway (ER)	Kolkata	1952
East Central Railway (ECR)	Hajipur	2002-03
East Coast Railway (ECoR)	Bhubaneswar	2002-03
Northern Railway (NR)	New Delhi	1952
North Central Railway (NCR)	Prayagraj	2002-03
North Eastern Railway (NER)	Gorakhpur	1952
Northeast Frontier Railway (NFR)	Guwahati (Maligaon)	1958
North Western Railway (NWR)	Jaipur	2002-03
Southern Railway (SR)	Chennai	1951
South Central Railway (SCR)	Secunderabad	1966
South Eastern Railway (SER)	Kolkata (Garden Reach)	1955
South East Central Railway (SECR)	Bilaspur	2002-03
South Western Railway (SWR)	Hubballi	2002-03
Western Railway (WR)	Mumbai (Churchgate)	1951
West Central Railway (WCR)	Jabalpur	2002-03
Metro Railway, Kolkata	Kolkata	2010 (as a zone)
South Coast Railway (SCoR)	Visakhapatnam	Announced 2019

Zone formation years are the commonly quoted dates; the newest zones came into effect in stages, so a few sources differ by a few months.

5. Train categories (premium to local)

Train type decides speed, stops, fare and the coaches attached. Below is a ladder from the fastest premium services down to slow local trains.

Category	Since	What it is
Vande Bharat Express	2019	Self-propelled, chair car (some sleeper versions), up to 160 km/h in service; fully air-conditioned; few stops
Rajdhani Express	1969	AC-only, connects state capitals to New Delhi; sleeper-style AC coaches; very few halts; meals included in fare
Shatabdi Express	1988	Day-time AC chair-car trains between cities a few hundred km apart; meals included
Duronto Express	2009	Non-stop or very few stops; long-distance point to point; AC and non-AC classes
Gatimaan Express	2016	Fast AC chair-car service on the New Delhi to Agra corridor, up to 160 km/h
Tejas Express	2017	Premium AC trains with modern interiors; some are run by IRCTC
Humsafar Express	2016	All 3-tier AC long-distance trains at simpler fares
Uday Express	2017	Double-decker AC chair-car trains
Garib Rath	2006	Low-fare AC trains (3-tier AC economy style coaches)
Jan Shatabdi	2002	Non-AC or economy version of Shatabdi for shorter trips
Sampark Kranti / Yuva	2000s	Superfast link trains to the capital; Yuva Express aimed at young travellers
Antyodaya / Amrit Bharat	2017 / 2023	Fully unreserved or non-AC push-pull trains for low-cost long-distance travel
Superfast Express	-	Average speed at least 55 km/h on broad gauge; supplementary fare
Mail / Express	-	Long-distance trains stopping at major stations; historically carried mail
Intercity Express	-	Day trains linking cities in a region
Fast Passenger / Passenger	-	Stop at most stations; unreserved; lowest fares
MEMU / DEMU	-	Mainline Electric or Diesel Multiple Unit trains for regional and suburban routes
EMU (suburban / local)	-	Urban commuter trains in Mumbai, Kolkata, Chennai, Delhi and other metro areas
Shuttle / toy trains	-	Short shuttles and hill heritage trains

Rule of thumb. Premium trains (Vande Bharat, Rajdhani, Shatabdi, Duronto, Tejas) make few stops and cost more; some use flexi or dynamic fares. Mail/Express and Superfast are the standard long-distance trains. Passenger and MEMU trains are for short trips and are usually unreserved.

6. Luxury, tourist and special trains

Train	Started	Notes
Palace on Wheels	1982	Rajasthan circuit, originally modelled on maharajas' saloon coaches
Royal Rajasthan on Wheels	2009	Weekly luxury tour of Rajasthan and Agra
Deccan Odyssey	2004	Luxury tourist train from Maharashtra covering western India
Golden Chariot	2008	South India luxury train, Karnataka and neighbouring states
Maharajas' Express	2010	Premium luxury train on several routes across India
Fairy Queen	1855 loco	Oldest working steam locomotive; heritage runs from Delhi area
Lifeline Express	1991	First hospital on rails; brings surgery and screening camps to rural areas
Bharat Gaurav trains	2021	Theme-based tourist circuits run through IRCTC and operators
Samjhauta / Thar / Maitree / Bandhan	1976 / 2006 / 2008 / 2017	Cross-border services to Pakistan and Bangladesh, several suspended over time

Special categories also include **relief and accident-relief trains**, **parcel and freight** services (goods, container and auto-carriers), **rail-cum-road** ferry links and **festival and holiday specials** that use the 0-series numbers.

7. Train numbering system

Train numbers have five digits. The first digit tells the service family; the pair of trains that make up one service usually differ by one digit and run in opposite directions. Since 2010 all trains use five digits.

First digit	Meaning
0	Special trains: festival, holiday and summer specials
1	Long-distance Mail/Express trains (for example 12xxx Superfast, 15xxx Express)
2	Long-distance Superfast and premium trains, including many Rajdhani, Shatabdi, Duronto and Vande Bharat numbers
3	Kolkata suburban trains
4	Suburban trains of other metro areas such as Delhi, Chennai, Secunderabad and Bengaluru
5	Passenger trains with conventional (diesel or electric) locomotives
6	MEMU (electric multiple unit) trains
7	DEMU (diesel multiple unit) trains
8	Special trains with a limited run
9	Mumbai suburban trains

Examples: **12951/12952** is the Mumbai to New Delhi Rajdhani pair; **20833** is a Vande Bharat number; **22xxx** is used for many superfast and Vande Bharat services. The exact sub-range depends on zone and category, so the first digit is a guide, not a rule.

8. Coach classes

The class code appears on the ticket and on search results. Codes below are those used in IRCTC and in this assistant.

Code	Class	Description
1A	AC First Class	Lockable cabins for 2 or 4; most expensive sleeping class
2A	AC 2-Tier	Open bays of berths in 2 tiers plus side berths; curtains and reading lights
3A	AC 3-Tier	3 tiers in bays of 6 plus 2 side berths; most popular AC class
3E	AC 3-Tier Economy	Like 3A with more berths per coach and simpler fare; used on Garib Rath and Humsafar
SL	Sleeper Class	Non-AC 3-tier sleeper; the standard budget sleeping class
CC	AC Chair Car	Air-conditioned seating for day journeys (Shatabdi, Vande Bharat)
EC	Executive Chair Car	Wider seats and extra space in premium day trains
VC / EA	Vistadome / Anubhuti	Glass-roof or premium-cabin coaches on scenic and premium trains
2S	Second Sitting	Non-AC seating, reserved or unreserved
FC	First Class (non-AC)	Older non-AC premium class, now rare
GEN	General / Unreserved	Unreserved second class; no seat guarantee

Berth types: Lower (LB), Middle (MB), Upper (UB), Side Lower (SL) and Side Upper (SU). Lower berths are prioritised for senior citizens, women over 45 and pregnant women where available.

9. Reservation quotas and booking status

A quota is a reserved share of seats. The general quota is what most passengers use.

Code	Quota	Meaning
GN	General	Main quota for all passengers
TQ / PT	Tatkal / Premium Tatkal	Last-minute booking, opens one day before the journey; higher charge; Premium Tatkal fare varies with demand
LD	Ladies	Seats reserved for women travellers
SS	Senior Citizen / Lower Berth	Reserved lower berths for older passengers and women
HP	Divyangjan (physically handicapped)	Seats for passengers with disabilities
DF	Defence	Reserved for armed forces personnel
FT	Foreign Tourist	Reserved for foreign tourists
PH	Parliament House	Reserved for members and officials
HO	Head Quarters / High Official	Reserved for officials
PQ	Pooled Quota	Shared between stations along the route
RS / RL	Roadside / Remote Location	Small stations away from the origin
LB	Lower Berth	Preference berth allocation

Booking status codes

Code	Meaning
CNF	Confirmed; a coach and berth or seat number is allotted
RAC	Reservation Against Cancellation; travel guaranteed, a seat or half a berth shared until a berth opens
WL / GNWL	General waitlist; moves up as cancellations happen
RLWL / PQWL / TQWL	Remote location, pooled quota or Tatkal waitlist
RSWL	Roadside waitlist
CAN / REL	Cancelled or released

Chart timing. Charts are prepared a few hours before departure. Waitlisted tickets that are not confirmed by the final chart are cancelled and refunded under the refund rules.

10. Heritage railways and metros

Line / site	Year	Notes
Darjeeling Himalayan Railway	1881	UNESCO World Heritage Site (1999); 2 ft gauge toy train with steam locomotives
Nilgiri Mountain Railway	1908	UNESCO (2005); metre gauge rack-and-pinion line from Mettupalayam to Ooty
Kalka to Shimla Railway	1903	UNESCO (2008); 2 ft 6 in gauge, with over 100 tunnels
Chhatrapati Shivaji Maharaj Terminus	1888	UNESCO (2004); Victorian Gothic station in Mumbai (originally Victoria Terminus)
Matheran Hill Railway	1907	Neral to Matheran, 2 ft gauge, a car-free hill station
Kangra Valley Railway	1929	Pathankot to Joginder Nagar narrow-gauge scenic line
Kolkata Metro	1984	India's first metro rail; now a separate railway zone
Other metros	2002 onwards	Delhi, Chennai, Bengaluru, Hyderabad, Mumbai, Kochi and others; run by city metro corporations, not Indian Railways

11. Records and landmarks

Record	Detail
Longest run (distance)	Vivek Express, Dibrugarh to Kanyakumari, about 4,200 km and roughly 80 hours
Fastest in service	Vande Bharat and Gatimaan run up to 160 km/h on selected sections
Longest platform	Gorakhpur Junction, about 1,366 m
Longest rail-road bridge	Bogibeel Bridge, Assam, about 4.9 km, opened 2018
Highest rail arch bridge	Chenab Bridge, Jammu and Kashmir, about 359 m above the river, opened 2025
Longest rail tunnel	Tunnels on the Kashmir link, including a 12+ km tunnel; earlier record was Pir Panjal (about 11 km)
Oldest surviving station	Royapuram, Chennai (1856)
Longest station name	Sri Venkatanarasimharajuvaripeta (Tamil Nadu)
Shortest station name	Ib, Odisha (two letters)
Oldest working locomotive	Fairy Queen (built 1855)

Safety milestones include the indigenous **Kavach** train-protection system, track modernisation and the removal of most level crossings on trunk routes. Rail accidents remain a serious concern, and safety spending has grown in recent budgets.

12. A to Z of notable trains

A quick index of well-known train names by first letter. Where a year is shown it is the start year usually quoted.

Letter	Trains
A	Amrit Bharat Express (2023): non-AC push-pull long-distance service. Antyodaya Express (2017): unreserved long-distance.
B	Bharat Gaurav: theme tourist trains (2021). Brindavan Express: Chennai to Bengaluru day train.
C	Coromandel Express: Howrah (Kolkata) to Chennai. Charminar Express: Hyderabad to Chennai.
D	Deccan Queen (1930): Mumbai to Pune. Deccan Odyssey (2004). Durgam Cheruvu Express (2009).
E	Express and Superfast trains; Executive Chair Car services.
F	Fairy Queen (1855 loco). Flying Raneer: Mumbai to Surat day train. Frontier Mail (1928, now Golden Temple Mail).
G	Gatimaan Express (2016). Garib Rath (2006). Godavari Express: Visakhapatnam to Hyderabad (Secunderabad). Grand Trunk Express.
H	Humsafar Express (2016). Howrah Mail. Himsagar Express: Kanyakumari to Jammu Tawi.
I	Intercity Express: day trains between cities. Island Express: Kanyakumari to Bengaluru.
J	Jan Shatabdi (2002). Janata Express: low-fare long-distance.
K	Konkan Kanya: Mumbai to Goa. Karnataka Express. Kerala Express.
L	Lifeline Express (1991): hospital train.
M	Maharajas' Express (2010). Mumbai Rajdhani. Maitree Express (2008): Kolkata to Dhaka. Mahanagari Express.
N	Navjeevan Express. Nilgiri Express (Chennai to Mettupalayam).
P	Palace on Wheels (1982). Punjab Mail (1912). Pushpak Express.
R	Rajdhani Express (1969). Royal Rajasthan on Wheels (2009).
S	Shatabdi Express (1988). Samjhauta Express (1976). Sampark Kranti. Sealdah Durgam.
T	Tejas Express (2017). Thar Express (2006). Taj Express: Delhi to Jhansi via Agra. Telangana Express.
U	Uday Express (2017). Utkal Express. Udayan Express: Mumbai to Bengaluru.
V	Vande Bharat Express (2019). Vivek Express (2011): longest distance run. Visakha Express.
Y	Yuva Express: economy long-distance service. Yeshwantpur Durgam.

Letters not listed (Q, W, X, Z, O) have no widely known train names of their own; several trains have more than one name over time.

13. A to Z glossary

Term	Meaning
Berth	A bed in a coach: lower, middle, upper, side lower or side upper.
Bogie	Frame holding wheels and axles under a coach or wagon.
Broad gauge	1,676 mm track, the standard in India.
Chart	List of confirmed passengers per coach, prepared before departure.
Coach	One passenger carriage; a train is a rake of coaches.
Concession	Fare discount for groups such as senior citizens, students, patients and Divyangjan.
Division	Smaller unit inside a zone, run by a Divisional Railway Manager.
DMU / EMU / MEMU	Diesel or electric multiple unit: self-propelled trains with no separate engine.
Down / Up	Directions of running; usually 'Up' is towards the zone or capital hub.
Dynamic fare	Fare that rises with demand on some premium trains.
E-ticket	Ticket booked online on IRCTC or its app.
Flexi fare	Fare that changes as seats fill on selected premium trains.
Gauge	Distance between rails.
Halt	A scheduled stop, shown with arrival and departure time.
IRCTC	Indian Railway Catering and Tourism Corporation.
Junction	Station where routes meet or diverge.
Kavach	Indigenous automatic train-protection system.
Loco / locomotive	Engine: steam (retired), diesel (WDM, WDP, WDG) or electric (WAP, WAG).
Mail / Express	Long-distance trains with limited stops.
NTES	National Train Enquiry System, the official live running information site.
PNR	Passenger Name Record, a 10-digit number identifying a booking.
Pantry car	Coach that prepares and serves meals on long routes.
Platform	Raised area at a station where trains stop.
Quota	Reserved share of seats such as general, ladies or Tatkal.
RAC	Reservation Against Cancellation.
Rake	A set of coaches that runs together as one train.
Signal / interlocking	Systems that allow or stop movement to prevent collisions.
Superfast	Category of faster trains with a supplementary charge.
Tatkal	Last-minute booking scheme, opens one day before travel.
Terminus	Station where a line ends.
Vande Bharat	Modern self-propelled fast train family launched in 2019.
Waitlist (WL)	Queue for a seat when a train is full.
Yard	Area of tracks used for sorting and storing wagons.
Zone	Top-level regional unit of Indian Railways.

14. Sources and how to verify

This guide was written from general, publicly available knowledge about Indian Railways. Numbers change often, so use the official sources below to confirm anything you plan to rely on.

- **Indian Railways** (indianrailways.gov.in): zones, divisions, policies, annual reports.
- **Ministry of Railways / Railway Board** circulars: current train numbers, fares and rules.
- **IRCTC** (irctc.co.in): booking, quotas, classes and current fares.
- **NTES** (enquiry.indianrail.gov.in): running status and schedules.
- **RDSO** and **Indian Railways Year Books**: technical and statistical detail.
- **UNESCO World Heritage List**: the four heritage railway entries above.

Accuracy note. Dates for early lines and zone formation are the commonly quoted ones and some sources differ. Records (longest, highest, fastest) are current as of this guide and can be beaten. For live train data use the Railway AI Assistant, which reads live status, schedules and PNR from provider APIs.